Modeling, Identification and Control of Mechatronic Systems
FALL 2025

OUT: 10/21/25 DUE: 11/04/25

Use separate sheets to answer each problem with clear problem number, figures, equations and illustrations. Please **submit the answer sheets** together with **Matlab codes and Simulink model** (Matlab2022a Version or below) used in your problem analysis when you hand in your homework. Please use the framework of Matlab codes (Rotary_Pendulum_Control.mlx) for reference.

PROBLEM #1 (System control design)

See the rotary inverted pendulum system in Homework 2, let the output of the system be the rod angle, i.e., $\alpha(t)$, the linearized dynamics of the rotary inverted pendulum system have the following structure with nonzero initial rod angle (for example, let $\alpha(t{=}0) = 6^{o}$):

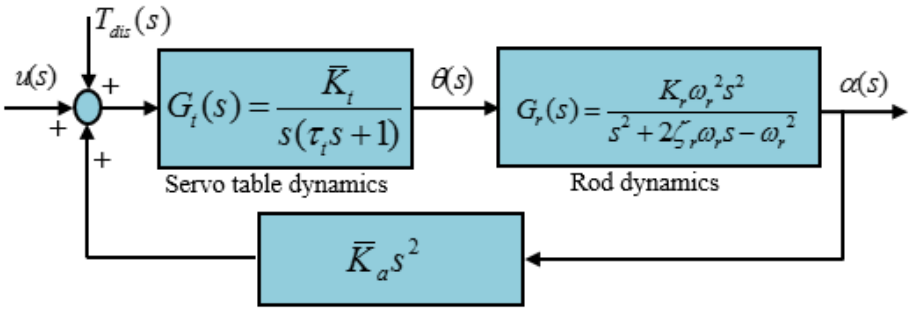

Figure 3.1

or the following equivalent I/O model:

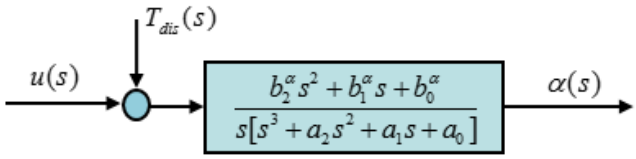

Figure 3.2 Plant model

Where, $a_0 = -\dfrac{\omega_t \omega_r^2}{1 - \bar{K}_a \bar{K}_t K_r \omega_t \omega_r^2}$, $a_1 = \dfrac{2\zeta_r \omega_t \omega_r - \omega_r^2}{1 - \bar{K}_a \bar{K}_t K_r \omega_t \omega_r^2}$, $a_2 = \dfrac{\omega_t + 2\zeta_r \omega_r}{1 - \bar{K}_a \bar{K}_t K_r \omega_t \omega_r^2}$, $b_2^\alpha = \dfrac{\bar{K}_t K_r \omega_t \omega_r^2}{1 - \bar{K}_a \bar{K}_t K_r \omega_t \omega_r^2}$,

$b_0^\alpha = b_1^\alpha = 0$, $\omega_t = \dfrac{1}{\tau_t}$, $\zeta_r = \sqrt{\dfrac{B_p^2}{4 m_p L_c I_p g}}$, $\omega_r = \sqrt{\dfrac{m_p L_c g}{I_p}}$, $K_r = \dfrac{L_t}{g}$, $\tau_t = \dfrac{I_t}{B_t}$, $\bar{K}_t = \dfrac{K_u}{B_t}$, $\bar{K}_a = \dfrac{m_p L_t L_c}{K_u}$,

$I_p = J_p + m_p L_c^2$, $I_t = J_t + m_p L_t^2$, $J_t = \eta_g K_g^2 J_m + J_l$, $K_u = \dfrac{\eta_m \eta_g k_t K_g}{R_m}$, $B_t = \dfrac{\eta_g \eta_m k_t k_m K_g^2 + B_{eq} R_m}{R_m}$,

$B_{eq} = B_r = \eta_g K_g^2 B_m + B_l$

You are required to synthesize controllers C(s) to meet the following performance requirements:

(i) Zero steady-state error for constant friction torque of rotary arm.

(ii) 2% settling time of the closed-loop system should be less than 0.5sec.

(iii) OS% should be no more than 20%

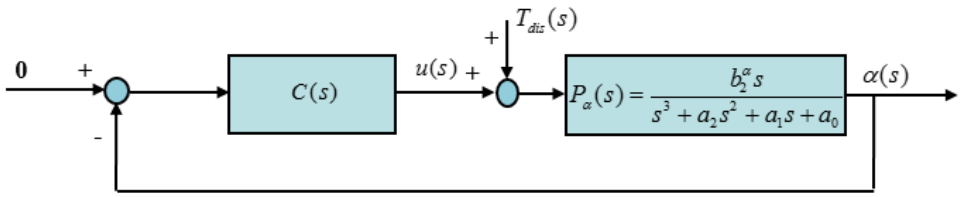

Figure 3.3 Block Diagram of RIP Control

To this end,

(A) Draw the transient performance region and address the requirement of controller structure to fulfill the three performance requirement?

(B) Let C(s) be a proper PID, i.e., $C(s) = K_p + \frac{K_I}{s} + \frac{K_d}{\tau_s + 1} s$. Using **pole-placement design** with placing the dominant CL poles to be at -10±10j and other CL poles to be -50, synthesize a controller that is expected to satisfy the three performance requirements. Obtain the corresponding control input, table angle and rod angle of the resulting closed-loop system with MATLAB. Check to see if the above performance is satisfied. If not, why?

(C) Let $C(s) = \frac{b_{c3}s^3 + b_{c2}s^2 + b_{c1}s + b_{c0}}{s^2(s + a_{c0})}$. Using **pole-placement design** with placing the dominant poles at appropriate places (the dominate CL poles are set as required in Table 3.1 and the other CL poles are set to be 5 times of CL poles), synthesize the controller that is expected to satisfy the three performance requirements. Obtain the corresponding control input, table angle and rod angle of the resulting closed-loop system under the following cases (Table 3.1) in MATLAB/Simulink simulation. What are the control performance influenced by the initial values of rod angle and closed-loop bandwidth?

Hint: performance requirements are used to place the desired poles. From (i), the denominator of the controller must include one integral part. From (ii), the desired damping ratio ζ_{cl} *can be specified according to* $\%OS = e^{\frac{-\pi\zeta_{cl}}{\sqrt{1-\zeta_{cl}^2}}} = e^{\frac{-\pi}{\tan(\varphi)}}$ *and* $\varphi = \tan^{-1}\left(\frac{\sqrt{1-\zeta_{cl}^2}}{\zeta_{cl}}\right)$. *From (iii), the desired natural frequency without damping is specified according to* $t_s \le 4\tau = \frac{4}{\sigma} = \frac{4}{\zeta_{cl}\omega_{cl}}$. *Then the dominate desired poles can be located at* $p_{d1,2} = -\zeta_{cl}\omega_{cl} \pm \omega_{cl}\sqrt{(\zeta_{cl}^2 - 1)}$.

Table 3.1 Cases for controller design with pole placement method

	Closed-loop bandwidth	Initial rod angle
Case 1	ω_{cl}=5,ζ_{cl}=0.707	α_0=6°
Case 2	ω_{cl}=15,ζ_{cl}=0.707	α_0=6°
Case 3	ω_{cl}=25,ζ_{cl}=0.707	α_0=6°
Case 4	ω_{cl}=5,ζ_{cl}=0.707	α_0=10°
Case 5	ω_{cl}=15,ζ_{cl}=0.707	α_0=10°
Case 6	ω_{cl}=25,ζ_{cl}=0.707	α_0=10°

Table 3.2 Parameters of the inverted pendulum

1 mbol	Description	Value	Unit
m_p	Mass of pendulum	0.127	kg

L_c	Half of the total length of pendulum	0.1685	m
J_p	Pendulum moment of intertia about center of mass	0.0012	kg.m^2
B_p	viscous damping coefficient of the pendulum	0.0024	N.m.s/rad
L_r	Rotary arm length from pivot to tip	0.216	m
J_r	Rotary arm moment of inertia about its center of mass	0.001	kg.m^2
B_r	Rotary arm viscous damping coefficient as seen at the pivot axis	0.0024	N.m.s/rad
R_m	Motor armature resistance	2.6	Ω
k_t	Motor current-torque constant	7.68×10^{-3}	N.m/A
k_m	Motor back-EMF constant	7.68×10^{-3}	V/(rad/s)
K_g	Total gear ratio	70	
η_m	Motor efficiency	0.69	
η_g	Gearbox efficiency	0.9	
A_c	Coulomb friction coefficient of input disturbance	0.4	
V_m	Applied voltage to the motor	-10~10	V